FundTrail — Screen Inventory and Navigation Map

Product	**FundTrail — personal finance management for Android**
Document	Docs/ux/screens.md
Version / status	1.2 — Draft for team review
Date	2026-10-21
Issue	EPIC-002 / T1 · Parent: #9
Built on	Docs/srs.md v1.4 (FR, NFR, CON, BR and D identifiers are cited as defined there)
Consumed by	EPIC-006 / T3 — the NavHost is built against section 5 of this document

How to use this document. Every screen has a stable ID. SCR-xx is a full-screen destination. DLG-xx is a dialog or bottom sheet that sits on top of a screen. IDs are never reused or renumbered; a dropped screen is marked ~~struck~~ (withdrawn). Each screen lists the FRs it serves, so a requirement can be traced to the screen that satisfies it (section 6 checks this in the other direction).

Terminology. The SRS calls the home screen the "dashboard" (FR-90…FR-99) and names its section "Insights". This document uses **Insights** for the screen and the tab label.

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1. Navigation principles

Principle	Source
Single activity. One NavHost using Navigation Compose with type-safe (serializable) routes and typed arguments.	CON-03
Screens never call the data layer. They read state from a ViewModel and send events to it. Navigation is triggered by UI events, not by repositories.	CON-04
Recording an expense is one tap from Insights (a FAB) and at most three taps in total.	FR-30, NFR-01
The app opens on Insights when a session exists. There is no splash screen of our own; the Android system splash covers cold start.	FR-06, FR-90
Onboarding is skippable and always ends on a working Insights screen.	FR-08
Back from a non-start tab returns to Insights. Back on Insights leaves the app.	Material navigation guidance
No screen shows streaks, missed-day counters or "incomplete" states.	FR-106
Every screen has an empty state, an actionable error with retry, and inline form validation.	FR-111, FR-112, FR-113

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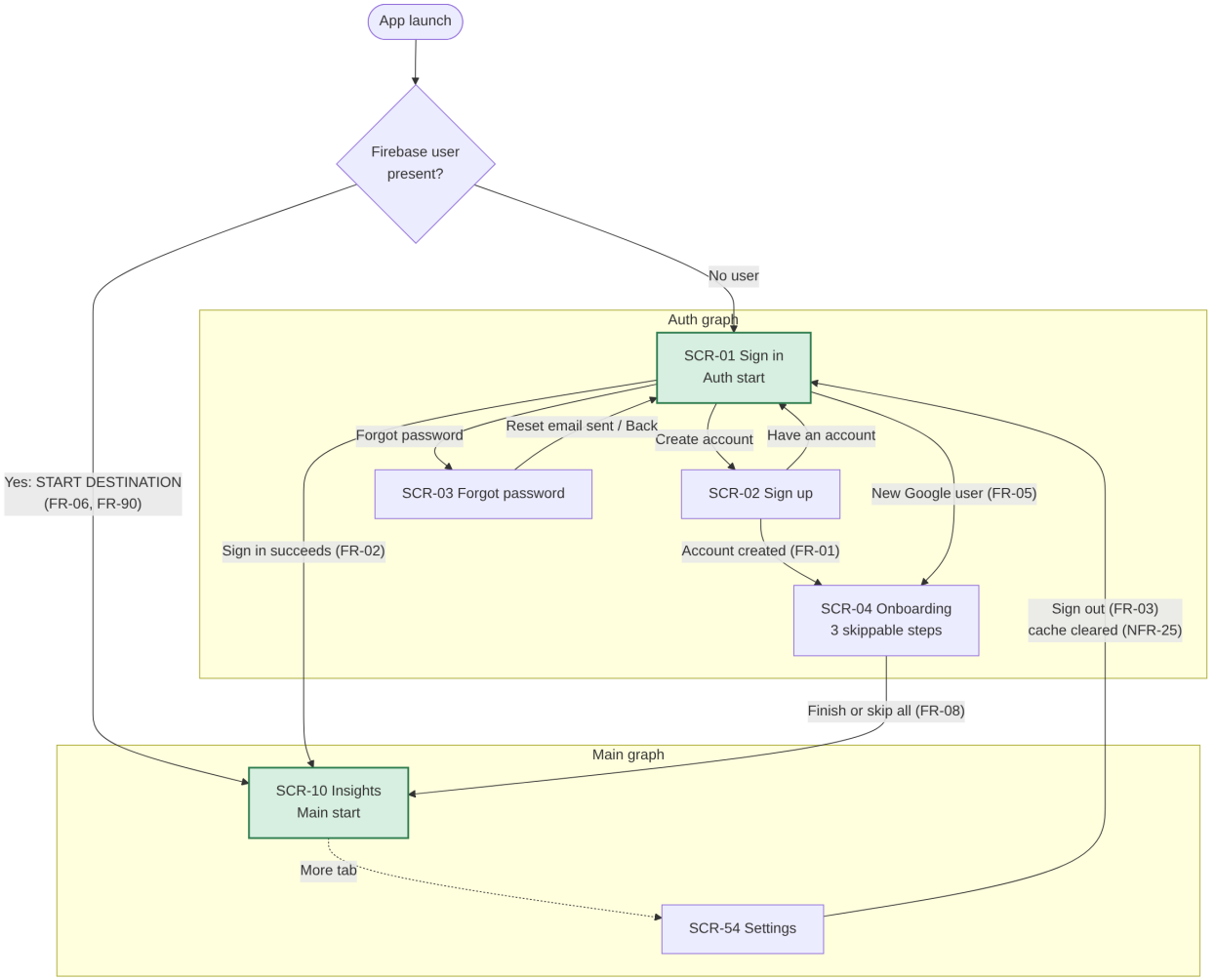
2. Structure at a glance

Item	Decision
Start destination (app)	A gate evaluated in `MainActivity` before the `NavHost` is composed: **no Firebase user → Auth graph** (starts at SCR-01 Sign in); **user present → Main graph** (starts at SCR-10 Insights).
Start destination (Auth graph)	SCR-01 Sign in
Start destination (Main graph)	SCR-10 Insights
Bottom-nav destinations (5)	SCR-10 Insights · SCR-11 History · SCR-12 Income · SCR-13 Goal · SCR-14 More
Where the bottom bar shows	Only on the five tab screens. It is hidden on entry forms, detail screens and dialogs.
Add expense	FAB on SCR-10 Insights (and a welcome-back action), opening SCR-20 with the amount field focused.
Leaving the Auth graph	`popUpTo(Auth) { inclusive = true }`, so Back from Insights never returns to sign-in.
Sign-out	From SCR-54 Settings. If writes haven't synced, DLG-10 asks for confirmation first (architecture.md §4.7). Then sign out and clear local data (NFR-25), `popUpTo(Main) { inclusive = true }` and navigate to SCR-01.
Tab switching	`launchSingleTop`, `restoreState`, `saveState`, popping to the Main graph's start destination. Each tab keeps its own back stack.
Deep links (notifications)	Daily reminder → SCR-20 (FR-40). Overdue milestone → SCR-32 for that project (FR-53).

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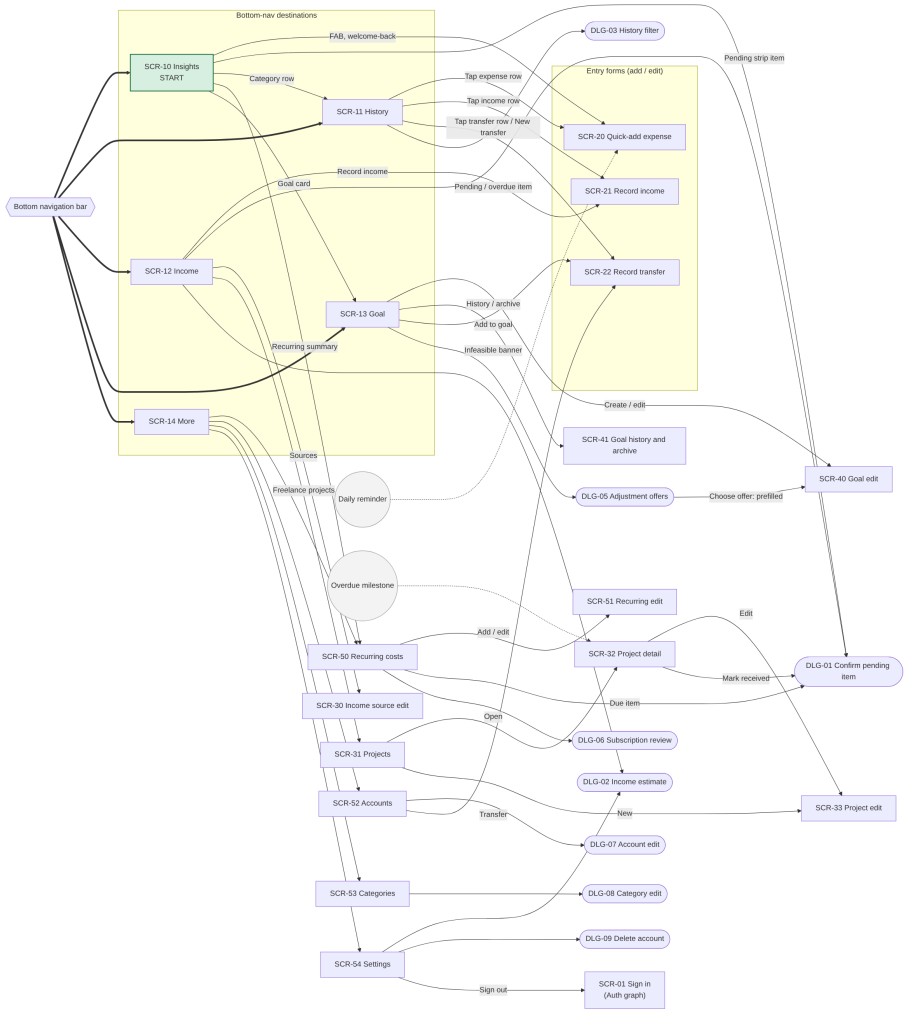
3. Navigation diagrams

3.1 Auth flow and start destination



3.2 Main graph, bottom navigation and drill-downs

Bold arrows are the bottom-navigation bar. Rounded boxes are dialogs and sheets. Entry screens return to the screen that opened them.



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4. Screen inventory

Type: *Tab* = bottom-nav destination · *Screen* = full-screen destination · *Dialog* = dialog or bottom sheet. Where an FR is shared by two screens, the one that owns the main interaction is listed first in section 6.

4.1 Authentication and onboarding (Auth graph)

ID	Screen	Type	Purpose	Main content and actions	FRs served
SCR-01	Sign in	Screen (Auth start)	Let a returning user into their data.	Email and password fields; Sign in; Google button (Could); links to Sign up and Forgot password. A wrong password shows an error and keeps the email entered.	FR-02, FR-05, FR-111, FR-113
SCR-02	Sign up	Screen	Create a new account.	Email, password, confirm password; Create account; Google button; link to Sign in. Duplicate or invalid email shows a specific message. Success opens SCR-04.	FR-01, FR-05, FR-113
SCR-03	Forgot password	Screen	Request a password-reset email.	Email field; Send reset email; confirmation message; Back to Sign in.	FR-04, FR-113
SCR-04	Onboarding	Screen	Capture the minimum setup in at most three steps, each skippable.	**Step 1 Goal:** name, target, deadline, and a linked savings account (pick one, or create one with its current balance as the opening balance). **Step 2 Accounts:** the five defaults, rename or archive. **Step 3 Income:** monthly estimate (low–high) and the four suggested sources. Skip on every step; Finish always lands on SCR-10.	FR-08, FR-10, FR-11, FR-45, FR-46, FR-58, FR-75

4.2 Bottom-nav destinations (Main graph)

ID	Screen	Type	Purpose	Main content and actions	FRs served
SCR-10	Insights	Tab (Main start)	Answer "where do I stand?" the moment the app opens.	Month switcher; summary (received, spent, saved); goal progress card (required vs actual saving, status, gap); committed vs discretionary split; ranked category breakdown; recurring-costs summary; pending strip (expected salary, overdue milestones, due recurring items, one tap to confirm); app-wide offline banner while offline (FR-110); FAB for quick-add; neutral welcome-back banner after 7 or more days with no entry. Useful with a single entry.	FR-30, FR-56, FR-90, FR-91, FR-92, FR-93, FR-94, FR-95, FR-97, FR-98, FR-99, FR-105
SCR-11	History	Tab	Find, check and correct any past record.	All transactions newest first, grouped by day, paged by 50; rows show type, amount, category or source, and account; transfers have a distinct style. Search by note or merchant; filter button opens DLG-03. Tap a row to edit; delete with an undo snackbar. Action to add a transfer.	FR-37, FR-61, FR-100, FR-101, FR-102
SCR-12	Income	Tab	Show what has actually arrived against what is expected.	This-month received, trailing 3-month average and 6-month range; the user's own estimate beside the computed one; income by source for the selected month; crypto year-to-date net; pending, expected and overdue items kept separate from received; entry points to sources, freelance projects and Record income.	FR-45, FR-49, FR-54, FR-55, FR-56, FR-57, FR-58, FR-59
SCR-13	Goal	Tab	Show feasibility of the one active goal, with no judgement.	Goal card: target, saved, deadline; required monthly saving; saving pace; status (ahead, on track, behind, or needs adjusting while the goal is infeasible, BR-09) with the LKR-per-month gap; projected completion date, or "no projection yet" when pace is zero or less; infeasible banner with adjustment offers; what-if preview (type a different monthly amount and see the projected date and status move, nothing saved); Add to goal (a transfer into the linked savings account, opens SCR-22 with it prefilled); crypto include/exclude toggle; Mark reached; links to Edit and History.	FR-75, FR-76, FR-77, FR-78, FR-79, FR-80, FR-81, FR-83, FR-84, FR-85
SCR-14	More	Tab	Hold the less frequent areas so the bar stays at five items.	List: Recurring costs, Accounts, Categories, Settings.	none directly (navigation hub)

4.3 Entry forms and confirmation

ID	Screen	Type	Purpose	Main content and actions	FRs served
SCR-20	Quick-add expense	Screen (add and edit)	Log a spend in seconds.	Amount focused with numeric keypad; category chips ordered by recent use, with the suggestion pre-selected and one tap to change it; date and time (defaults to now, back-dating allowed); account (defaults to last used); optional note or merchant; optional expense-type override; optional foreign-currency block (original amount, LKR value, derived rate). Save; Save and add another (keeps date and account). Only the amount is required, so a missing category saves as Uncategorised. On saving a discretionary expense, a neutral goal-cost message appears.	FR-22, FR-23, FR-24, FR-25, FR-30, FR-31, FR-32, FR-33, FR-35, FR-36, FR-37, FR-38, FR-41, FR-113
SCR-21	Record income	Screen (add and edit)	Record received money from any source.	Source picker; original currency and amount; LKR received; derived rate (never recomputed); date; account. Crypto sources take a signed gain or loss. AdSense can be saved as `EXPECTED` with an estimated LKR amount, which counts as zero until confirmed as `ACTUAL`.	FR-45, FR-47, FR-54, FR-55, FR-113
SCR-22	Record transfer	Screen (add and edit)	Move money between the user's own accounts without distorting totals.	From account, to account, amount, date. Same from and to is rejected with a message. Moves into or out of Binance use this form.	FR-60, FR-61, FR-62, FR-113
DLG-01	Confirm pending item	Dialog	Confirm an `EXPECTED` item as `ACTUAL` with one confirmation.	Shows the expected amount; the actual amount is editable; date and account; shows the variance (actual − expected). Used for expected salary, an `EXPECTED` AdSense entry, a freelance milestone marked received (linked to its project and milestone) and a due recurring item.	FR-49, FR-52, FR-54, FR-66, FR-67, FR-99
DLG-03	History filter	Bottom sheet (in-screen, UX-07)	Narrow the history list.	Type, account, category, source, date range; Apply and Clear.	FR-101

4.4 Income area

ID	Screen	Type	Purpose	Main content and actions	FRs served
SCR-30	Income source edit	Screen (add and edit)	Define where money comes from.	Name, type (salary, freelance, AdSense, crypto, other), cadence, currency, optional expected amount, optional expected day.	FR-45
SCR-31	Freelance projects	Screen	List projects and their payment status.	Filter tabs (all, pending, overdue, received); projects with total agreed and a status chip per milestone (pending, overdue, received); New project.	FR-50, FR-51, FR-56
SCR-32	Project detail	Screen	Track one project's milestones and chase unpaid ones.	Project name, client, total agreed; milestones with amount, due date and status; Mark received (opens DLG-01, creates the linked income entry); Edit. Target of the overdue-milestone notification.	FR-50, FR-51, FR-52, FR-53
SCR-33	Project edit	Screen (add and edit)	Create or change a project and its milestones.	Name, optional client, total agreed, milestone rows (amount, due date); add and remove rows.	FR-50
DLG-02	Income estimate	Dialog	Let the user state their own expected monthly income.	Low and high values, shown beside the computed figure on SCR-12. Opened from SCR-12 and SCR-54.	FR-58, FR-115

4.5 Goal area

ID	Screen	Type	Purpose	Main content and actions	FRs served
SCR-40	Goal edit	Screen (create and edit)	Create the goal or change its target or deadline without losing progress.	Name, target (LKR), deadline, linked savings account; live preview of the required monthly saving. Accepts values prefilled from an adjustment offer. Saving an edit to target or deadline logs the old value.	FR-75, FR-76, FR-81, FR-82
SCR-41	Goal history and archive	Screen	Show how the goal changed and what has been reached.	Revision history (old and new target or deadline, date); archived goals; Create a new goal once the last one is archived.	FR-82, FR-84
~~DLG-04~~	~~Add contribution~~ (withdrawn)	—	Withdrawn in v1.1: adding to the goal is a transfer into its linked savings account (SRS v1.3 BR-07, FR-80), entered on SCR-22.	—	—
DLG-05	Goal adjustment offers	Bottom sheet (in-screen, UX-07)	Present the two adjustments when the goal is infeasible.	(a) new deadline; (b) reduced target; Edit manually; Dismiss. Choosing an offer opens SCR-40 prefilled.	FR-81

4.6 Recurring costs, accounts, categories and settings (reached from More)

ID	Screen	Type	Purpose	Main content and actions	FRs served
SCR-50	Recurring costs	Screen	See every recurring cost and confirm what is due.	Templates with expected amount, cadence, next due day and a combined monthly total; due items with one-tap confirm (DLG-01); history showing expected vs actual with the variance; pause or end a template; subscription-review prompts.	FR-65, FR-66, FR-67, FR-68, FR-69, FR-70
SCR-51	Recurring edit	Screen (add and edit)	Create or change a recurring template.	Name, category, expected amount, cadence (weekly, monthly, yearly), due day, account, type; Pause; End.	FR-65, FR-70
DLG-06	Subscription review	Dialog	Ask "still using this?" every 90 days, without pressure.	Keep, Remind me to cancel, End. *End* stops future prompts and keeps history.	FR-69
SCR-52	Accounts	Screen	Manage the user's pots and see each balance.	The five defaults plus any added; balance per account (opening balance + income − expenses ± transfers); add, rename, archive (archived accounts leave entry forms but keep history); Transfer action.	FR-10, FR-11, FR-12, FR-60
DLG-07	Account edit	Dialog	Add or rename an account.	Name, kind, optional opening balance, archive.	FR-11
SCR-53	Categories	Screen	Manage categories and their committed or discretionary type.	Seeded and custom categories grouped by type; change a category's type; add, rename, archive.	FR-20, FR-21, FR-22
DLG-08	Category edit	Dialog	Add or change one category.	Name, default type, archive.	FR-21, FR-22
SCR-54	Settings	Screen	Hold preferences and account-level actions.	Daily reminder time (off by default); income estimate (opens DLG-02); subscription-review interval; include crypto in goal projection (default off); export all records as CSV; delete account and data; sign out. The theme follows the system setting, so there is no theme switch.	FR-03, FR-40, FR-58, FR-69, FR-83, FR-115, FR-116, FR-118
DLG-09	Delete account	Dialog	Confirm the irreversible delete of the account and all its data.	Plain-language consequence; Cancel; Delete. After deletion the user returns to SCR-01.	FR-117

ID	Screen	Type	Purpose	Main content and actions	FRs served
DLG-10	Confirm sign-out	Dialog	Warn before signing out with changes that haven't synced.	Shown only when pending writes don't finish within a few seconds: "Some changes haven't synced yet and will be lost if you sign out." Cancel; Sign out anyway (architecture.md §4.7).	FR-03, NFR-25

4.7 Behaviour that belongs to no single screen

These FRs are met by shared components or by the data layer, so they appear in every relevant screen's design rather than as a destination.

Concern	FRs	How it is met in the UI
Session gate	FR-06	The start-destination gate in section 2.
Per-user data scope	FR-07	Data layer. No UI.
Default data seeding	FR-10, FR-20	First sign-in creates the five accounts and the Appendix A categories. They are then visible in SCR-52 and SCR-53.
Record structure and stored values	FR-34, FR-48	Data layer. The forms only supply the fields.
Offline entry	FR-39	Every form saves locally first and never blocks on the network.
Real-time updates	FR-96	SCR-10 collects a `StateFlow` built from Room queries. The sync listeners (one per subcollection, `data/sync`) keep Room up to date, so a saved entry shows up without refresh.
Offline banner	FR-110	One app-wide banner in the main scaffold, driven by connectivity (architecture.md §4.6). Screens don't handle connectivity themselves.
Errors, empty states, validation	FR-111, FR-112, FR-113	Shared error component with Retry; every list and chart has an empty state; inline messages on every form.
Neutral tone, no streaks	FR-105, FR-106, NFR-14	Copy review across every screen. The welcome-back banner is part of SCR-10.
Overdue-milestone reminder	FR-53	A local notification that deep-links to SCR-32.

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5. Route table for the NavHost (EPIC-006 / T3)

Routes are `@Serializable` types. Optional arguments default to `null`; a `null` id means "create". Nested graphs are `Auth` and `Main`.

Graph	Route	Screen / dialog	Arguments	Entered from
Auth	`SignIn` (start)	SCR-01	none	Gate; Sign up; Forgot password; sign-out; delete account

Graph	Route	Screen / dialog	Arguments	Entered from
Auth	SignUp	SCR-02	none	SCR-01
Auth	ForgotPassword	SCR-03	none	SCR-01
Auth	Onboarding	SCR-04	none	SCR-02; new Google user at SCR-01
Main	Insights (start, tab)	SCR-10	none	Gate; SCR-01; SCR-04; bottom bar
Main	History (tab)	SCR-11	categoryId: String? = null	Bottom bar; SCR-10 category row
Main	Income (tab)	SCR-12	none	Bottom bar
Main	Goal (tab)	SCR-13	none	Bottom bar; SCR-10 goal card
Main	More (tab)	SCR-14	none	Bottom bar
Main	QuickAdd	SCR-20	transactionId: String? = null	SCR-10 FAB; SCR-11; reminder deep link
Main	RecordIncome	SCR-21	transactionId: String? = null, sourceId: String? = null	SCR-12; SCR-11
Main	RecordTransfer	SCR-22	transactionId: String? = null, toAccountId: String? = null	SCR-11; SCR-52; SCR-13 (Add to goal, prefills the savings account)
Main	ConfirmPending (dialog)	DLG-01	kind: PendingKind (SALARY, ADSENSE, MILESTONE, RECURRING), itemId: String	SCR-10 strip; SCR-12; SCR-32; SCR-50
Main	IncomeSourceEdit	SCR-30	sourceId: String? = null	SCR-12
Main	Projects	SCR-31	none	SCR-12
Main	ProjectDetail	SCR-32	projectId: String	SCR-31; overdue-milestone deep link
Main	ProjectEdit	SCR-33	projectId: String? = null	SCR-31; SCR-32
Main	IncomeEstimate (dialog)	DLG-02	none	SCR-12; SCR-54
Main	GoalEdit	SCR-40	goalId: String? = null, prefillTarget: Long? = null, prefillDeadline: Long? = null	SCR-13; DLG-05
Main	GoalHistory	SCR-41	none	SCR-13
Main	Recurring	SCR-50	none	SCR-14; SCR-10 recurring summary
Main	RecurringEdit	SCR-51	templateId: String? = null	SCR-50

Graph	Route	Screen / dialog	Arguments	Entered from
Main	SubscriptionReview (dialog)	DLG-06	templateId: String	SCR-50
Main	Accounts	SCR-52	none	SCR-14
Main	AccountEdit (dialog)	DLG-07	accountId: String? = null	SCR-52
Main	Categories	SCR-53	none	SCR-14
Main	CategoryEdit (dialog)	DLG-08	categoryId: String? = null	SCR-53
Main	Settings	SCR-54	none	SCR-14
Main	DeleteAccount (dialog)	DLG-09	none	SCR-54
Main	ConfirmSignOut (dialog)	DLG-10	none	SCR-54

Notes for the build:

- Money amounts passed as arguments are integer minor units (BR-02). Prefer ids over objects; ViewModels load the record by id.
- The bottom bar is shown when the current destination is one of the five tab routes, and hidden otherwise.
- DLG-03 and DLG-05 are bottom sheets shown by their screen, so they have no route (UX-07).
- All back-stack rules in section 2 apply. In particular, sign-out and delete-account clear the whole Main graph.

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6. Requirement coverage

Every functional requirement in the SRS maps to a screen in section 4 or to a shared behaviour in section 4.7.

SRS section	FRs	Served by
5.1 Authentication and onboarding	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08	SCR-02, SCR-01, SCR-54 and DLG-10, SCR-03, SCR-01/02, gate, data layer, SCR-04
5.2 Accounts	FR-10, FR-11, FR-12	SCR-52, DLG-07, SCR-04
5.3 Categories	FR-20, FR-21, FR-22, FR-23, FR-24, FR-25	SCR-53, DLG-08, SCR-20
5.4 Expense entry	FR-30, FR-31, FR-32, FR-33, FR-34, FR-35, FR-36, FR-37, FR-38, FR-39, FR-40, FR-41	SCR-20 (FR-37 also SCR-11; FR-40 in SCR-54; FR-34 and FR-39 cross-cutting)

SRS section	FRs	Served by
5.5 Income	FR-45, FR-46, FR-47, FR-48, FR-49, FR-50, FR-51, FR-52, FR-53, FR-54, FR-55, FR-56, FR-57, FR-58, FR-59	SCR-30, SCR-04, SCR-21, DLG-01, SCR-31/32/33, SCR-12, DLG-02
5.6 Transfers	FR-60, FR-61, FR-62	SCR-22 (FR-61 also SCR-11)
5.7 Recurring costs	FR-65, FR-66, FR-67, FR-68, FR-69, FR-70	SCR-50, SCR-51, DLG-01, DLG-06
5.8 Savings goal	FR-75, FR-76, FR-77, FR-78, FR-79, FR-80, FR-81, FR-82, FR-83, FR-84, FR-85	SCR-13, SCR-40, SCR-41, DLG-05 (FR-80 via SCR-22; FR-83 also SCR-54)
5.9 Insights	FR-90, FR-91, FR-92, FR-93, FR-94, FR-95, FR-96, FR-97, FR-98, FR-99	SCR-10
5.10 History	FR-100, FR-101, FR-102	SCR-11, DLG-03
5.11 Gap tolerance	FR-105, FR-106	SCR-10 banner; all screens
5.12 Reliability, errors and settings	FR-110, FR-111, FR-112, FR-113, FR-115, FR-116, FR-117, FR-118	Shared components (section 4.7); SCR-54, DLG-09

The use cases in SRS section 2.3 map to screens as follows.

UC	Primary screens
UC-01 Quick-add expense	SCR-10 → SCR-20
UC-02 Confirm salary received	SCR-10 strip or SCR-12 → DLG-01
UC-03 Track freelance milestone	SCR-12 → SCR-31 → SCR-32 → DLG-01
UC-04 Record AdSense payout	SCR-12 → SCR-21, or DLG-01 to confirm an estimate
UC-05 Record crypto result	SCR-12 → SCR-21
UC-06 Record transfer	SCR-11 or SCR-52 → SCR-22
UC-07 Check dashboard	SCR-10
UC-08 Manage MacBook goal	SCR-13 → SCR-40, SCR-22 (Add to goal), DLG-05
UC-09 Review recurring costs	SCR-14 → SCR-50 → DLG-01, DLG-06
UC-10 Catch up after a gap	SCR-10 banner → SCR-20 (Save and add another)
UC-11 Sign up / sign in	SCR-01, SCR-02, SCR-04

UC	Primary screens
UC-12 Manage accounts	SCR-14 → SCR-52 → DLG-07
UC-13 Manage categories	SCR-14 → SCR-53 → DLG-08
UC-14 Browse history	SCR-11 → DLG-03
UC-15 Manage settings and data	SCR-14 → SCR-54 → DLG-02, DLG-09

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7. Decisions and open points for review

ID	Point	Proposal	Why
UX-01	Where should the app decide between Auth and Main?	A gate before the `NavHost` reads the Firebase user and picks the start graph. No Splash destination.	Avoids a flash of the wrong screen and matches FR-06 "open straight to the dashboard".
UX-02	How many bottom-nav items?	Five: Insights, History, Income, Goal, More.	Material 3 allows three to five. Recurring, Accounts, Categories and Settings are used less often than the four daily areas.
UX-03	How are income and transfers entered, given that the FAB is for expenses?	The FAB opens quick-add only. Income is entered from SCR-12, transfers from SCR-11 and SCR-52.	Keeps NFR-01 (three taps for the most frequent action) without a speed-dial menu.
UX-04	Should Sign in resume an unfinished onboarding?	No. Onboarding follows account creation only (FR-01). A user who leaves it early lands on Insights with empty-state prompts (FR-08).	Keeps onboarding free of state and consistent with "reach the dashboard even if every step is skipped".
UX-05	Dialogs vs destinations	Small confirm and edit forms (DLG-xx) are `dialog` destinations, so they are deep-linkable and survive rotation. Bottom sheets are the exception (UX-07).	NFR-17 requires rotation not to crash.
UX-06	Is the tab layout needed for a tablet?	No, phones in portrait only.	SRS section 1.3 puts tablet layouts out of scope.
UX-07	How are bottom sheets navigated?	DLG-03 and DLG-05 are `ModalBottomSheet`s shown by their screen from ViewModel state, with no route.	Navigation Compose's `dialog()` destination can't host a bottom sheet, and these two are never deep-linked. Sheet visibility lives in the ViewModel, so it survives rotation (NFR-17).

Change log

Version	Date	Change
1.0	2026-10-07	First draft: screen inventory, auth and main navigation diagrams, route table and FR coverage.
1.1	2026-10-09	Review fixes (#65): built on SRS v1.3; offline banner on SCR-10; Add to goal is a transfer (DLG-04 and the `AddContribution` route withdrawn, `RecordTransfer` takes `toAccountId`); FR-85 what-if on SCR-13; `EXPECTED` / `ACTUAL` wording; SCR-40 uses the linked savings account; DLG-10 Confirm sign-out and its route; bottom sheets DLG-03 and DLG-05 are in-screen with no route (UX-07).
1.2	2026-10-21	M3 wireframe review (#124): SCR-31 gains the filter tabs drawn in `S4-income-invoices.md`; SCR-13 lists the *needs adjusting* status added by SRS v1.4 (BR-09, FR-78).